

Research Preparation Document: Public-Literature Anchors for the Universality Theorem for Foundation MLIPs

Purpose. This document is the second-paper research preparation companion to the planned trilogy whose first paper (CMET, in revision) provides the manifold-extension primitive, whose second paper (this manuscript, "Universality Theorem") asserts that every foundation MLIP in a defined architectural class F admits a per-model error manifold with class-uniform structural properties, and whose third paper will treat classical-potential refusal-mode behavior. The job of this document is to replace any internal-corpus citations with public-literature anchors — arXiv, peer-reviewed journals, NeurIPS/ICML/ICLR proceedings, and benchmark websites — and to be explicit about which mathematical primitives are direct citations, which require an extension, and which become pre-registration anchors for falsifiable predictions.

The document is organized in the five parts the manuscript itself will require. Throughout, the architectural class F is the set of foundation MLIPs satisfying (F1) pretraining on a chemically diverse DFT corpus at scale, (F2) representational expressivity strictly above the Pozdnyakov-Cerioti incompleteness floor, and (F3) smoothness of the model M as a function of atomic positions outside a measure-zero set of singular configurations.

PART 1: PUBLIC EMPIRICAL ANCHORS

The Universality Theorem requires empirical anchors for three structural facts: (1A) that foundation MLIPs trained on overlapping data show systematic error structure that varies substantially across models; (1B) that the error structure is low-dimensional in a sloppy-spectrum sense; and (1C) that there exist cross-paradigm directions in which multiple foundation MLIPs agree, indicating shared structural inadequacy rather than model-specific noise.

1A. Cross-model error heterogeneity on overlapping data

The single most important empirical anchor is the Matbench Discovery leaderboard. Riebesell, Goodall, Benner, Chiang, Deng et al. introduced the framework in arXiv:2308.14920 (2023), published as Riebesell et al., *Nature Machine Intelligence* 7, 836–847 (2025), DOI 10.1038/s42256-025-01055-1. As of the most recent leaderboard pull (matbench-discovery.materialsproject.org), the platform ranks 47 distinct models on a unified WBM test set with DFT reference convex hull, including foundation MLIPs from at least eleven distinct architectural families — MACE-MP-0, MACE-MPA-0, CHGNet, Orb, Orb-v3, SevenNet, EquiformerV2 (eqV2), GNoME, MatterSim, eSEN, GRACE, DPA3, PET-MAD, and UMA/OMat24-derived models. The leaderboard's headline finding for our purposes is heterogeneous error structure: the F1 scores for stability classification span more than a factor of three across compliant models, and the κ_{SRME} thermal-conductivity

benchmark (Póta, Ahlawat, Csányi, Simoncelli arXiv:2408.00755, 2024) produces a different ordering than the energy-MAE ordering. As Chiang et al. put it directly in MLIP Arena (arXiv:2509.20630, NeurIPS 2025 D&B Track), "many top-ranked models on bulk crystal EOS and Matbench Discovery perform poorly in pairwise interactions, suggesting that apparent benchmark success may result from plausible many-body error cancellation in bulk systems." This is exactly the cross-model error-structure heterogeneity the Universality Theorem clause (iii) (adaptivity) predicts. (arXiv)

MLIP Arena (Chiang, Kreiman, Zhang, Kuner et al., arXiv:2509.20630, 2025) is the second indispensable anchor, because it evaluates physics-aware properties — pairwise PEC accuracy, MD stability under NVT ramps from 300 K to 3000 K, chemical reactivity — that are orthogonal to energy/force MAE on near-equilibrium data and therefore expose model-specific failure modes. The Arena paper explicitly criticizes error-based DFT-reference benchmarks for "data leakage, limited transferability, and an over-reliance on error-based metrics tied to specific density functional theory (DFT) references" and finds that "non-compliant models rank highly for crystal stability metrics due to energy overfitting at the expense of forces and finite-temperature capabilities." The two-body PEC dispersion across MACE-MP-0, CHGNet, Orb-family, SevenNet, GRACE, eqV2, MatterSim, eSEN, and PET-MAD provides a direct per-element error-pattern fingerprint for each model. (arXiv) (arXiv)

The PET-MAD paper (Mazitov, Bigi, Kellner, Pegolo, Tisi, Fraux, Pozdnyakov, Loche, Ceriotti, *Nature Communications* 16, 10653 (2025), DOI 10.1038/s41467-025-65662-7; arXiv:2503.14118) is the third anchor because it explicitly trades dataset size for diversity and demonstrates that "recent 'universal' models deliver qualitative accuracy across the periodic table but are often biased toward low-energy configurations" — that is, that two MLIPs trained on near-equilibrium data share a low-energy bias direction that vanishes once the training distribution is broadened. This is direct evidence for the class-uniform component of the error manifold. (arxiv)

Cross-functional transferability evidence comes from Huang, Deng, Zhong, Kaplan, Persson, Ceder, arXiv:2504.05565 (2025), "Cross-functional transferability in universal machine learning interatomic potentials," which documents that GGA-trained CHGNet variants and r²SCAN-trained variants disagree on absolute energies but agree on the structure of the residual once elemental energy referencing is applied. This is a fourth empirical fingerprint of class-uniform versus model-specific error decomposition. (arXiv)

For molecular foundation MLIPs the analogous anchor is MACE-OFF23 (Kovács, Moore, Browning, Batatia, Horton, Kapil, Witt, Magdău, Cole, Csányi, arXiv:2312.15211, 2023; published JACS 2025, DOI 10.1021/jacs.4c07099). MACE-OFF23(L) achieves 0.5–1.0 meV/atom energy error and 15–20 meV/Å force error on drug-like organics, with intermolecular force RMSE of 5–15 meV/Å — figures the paper states as "the large model generally achieving errors of around 0.5–1.0 meV/atom and 15–20 meV/Å, well below the 1 kcal/mol (43 meV) chemical accuracy limit" and "MACE-OFF23(M) and MACE-OFF23(L) both yield similar errors of around 5–15 meV/Å" for intermolecular forces. These per-element and per-property residuals can be re-decomposed

across MACE-OFF23, AIMNet2, and ANI-2x to recover cross-MLIP correlation structure.

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MatterSim (Yang et al., arXiv:2405.04967, 2024) provides element-resolved error breakdowns explicitly: a mean absolute error of 36 meV/atom on the MPF-TP benchmark (which spans 0–5000 K and 0–1000 GPa), signifying, in the authors' own words, "an up to ten-fold enhancement in precision compared to the prior best-in-class," and "2 to 3-fold more distinct atomic environments across the entire periodic table compared to previous databases... and 10-fold or even higher for certain elements." For the manuscript, MatterSim is the cleanest single example of a foundation MLIP whose error covariance has both a model-specific direction (architectural choice) and a class-uniform direction (training-distribution coverage). (arxiv + 2)

The Loew/Bochkarev/Drautz lineage deserves a careful citation: the arXiv ID 2503.18913 originally referenced does not point to an MLIP paper (it is an atomic-physics paper on argon ionization). The substantive cross-MLIP-comparison reference in this lineage that I will use in the manuscript is Loew, Sun, Wang, Botti, Marques, "Universal machine learning interatomic potentials are ready for phonons," arXiv:2412.16551, *npj Computational Materials* 11, 178 (2025), DOI 10.1038/s41524-025-01650-1, which compares MACE-MP-0, CHGNet, M3GNet, SevenNet, Orb, eqV2, GRACE-2L on phonon-property prediction across the MDR phonon benchmark and finds that conservative MLIPs systematically outperform direct-force models, again with model-specific error structure. The closely related GRACE architecture anchor is Bochkarev, Lysogorskiy, Drautz, *Phys. Rev. X* 14, 021036 (2024), "Graph atomic cluster expansion for semilocal interactions beyond equivariant message passing." (arXiv + 2)

Surface-property failure modes — relevant to clause (v) refusal — are documented by Focassio, Mezzina Freitas, Schleder, *ACS Applied Materials & Interfaces* 17, 13111–13121 (2025), DOI 10.1021/acsami.4c03815 (preprint arXiv:2403.04217): MACE, CHGNet, and M3GNet "have significant shortcomings in this task, with errors correlated to the total energy of surface simulations, having an out-of-domain distance from the training dataset." This is the cleanest published example of an out-of-distribution failure that the refusal-mode clause (v) is designed to predict. (arXiv) (arxiv)

The headline finding for clause (iii) "systematic softening" comes from Deng, Choi, Zhong, Riebesell, Anand, Li, Jun, Persson, Ceder, *npj Computational Materials* 11, 9 (2025), DOI 10.1038/s41524-024-01500-6 (preprint arXiv:2405.07105). The paper documents "a consistent potential energy surface (PES) softening effect in three uMLIPs: M3GNet, CHGNet, and MACE-MP-0, which is characterized by energy and force underprediction in atomic-modeling benchmarks including surfaces, defects, solid-solution energetics, ion migration barriers, phonon vibration modes, and general high-energy states... The PES softening behavior originates primarily from the systematically underpredicted PES curvature, which derives from the biased sampling of near-equilibrium atomic arrangements in uMLIP pre-training datasets." Three foundation MLIPs with different architectures share the same direction of systematic error

— this is the most quantitatively unambiguous cross-paradigm cosine-alignment finding currently in the public literature. (eScholarship) (Nature)

The Christiansen–Hammer Δ -correction paper (Christiansen and Hammer, arXiv:2502.21179, 2025) reinforces the cross-paradigm finding from a different angle: "Other universal potentials trained on the same data, MACE-MP0, SevenNet-0, and ORB-v2-only-MPtrj show similar behavior, but with varying degrees of error, demonstrating the general need for augmentation schemes for universal potential models." This is the empirical license for the manuscript's Δ -learning corollary and for the embedding-space residual framework. (arxiv)

1B. Low-dimensional sloppy-spectrum error structure for MLIPs

Direct evidence that an MLIP-specific Fisher information matrix exhibits geometric eigenvalue decay (the sloppy-spectrum signature) is thinner than 1A. The most direct citation is the Kurniawan, Petrie, Williams, Transtrum, Tadmor, Elliott, Karls, Wen paper, "Bayesian, Frequentist, and Information Geometry approaches to parametric uncertainty quantification of classical empirical interatomic potentials" (Transtrum group lineage, the work in the DUNN / OpenKIM line) — explicit sloppy-eigenvalue analysis for classical interatomic potentials, the prototype of clause (i). For neural-network interatomic potentials, the strongest current citation is Perez, Subramanyam, Maliyov, Swinburne, *npj Computational Materials* 11, 263 (2025), arXiv:2502.07104, which demonstrates that for misspecified MLIPs the relevant uncertainty is a parameter-space covariance with strongly anisotropic spectrum and gives explicit propagation to property predictions. The companion paper Swinburne and Perez, *Machine Learning: Sci. Technol.* 6, 015008 (2025), DOI 10.1088/2632-2153/ad9fce, treats parameter uncertainties for imperfect surrogate models in the low-noise regime — again with explicit sloppy structure.

(BYU Physics) (arXiv)

The neural-network-Hessian and Fisher-information lineage that supports the sloppy structure at the broader theoretical level comprises four anchor papers. (i) Sagun, Evci, Güney, Dauphin, Bottou, "Empirical Analysis of the Hessian of Over-Parametrized Neural Networks," arXiv:1706.04454 (preprint, 2017), found empirically that "the spectrum of the Hessian is composed of two parts: (1) the bulk centered near zero, (2) and outliers away from the bulk... Fixing data, increasing the number of parameters merely scales the bulk of the spectrum." (ii) Karakida, Akaho, Amari, "Universal statistics of Fisher information in deep neural networks: mean field approach," AISTATS 2019 (PMLR Vol. 89, 1032–1041), arXiv:1806.01316, prove that "most of [the FIM's eigenvalues] are close to zero while the maximum eigenvalue takes a huge value. Because the landscape of the parameter space is defined by the FIM, it is locally flat in most dimensions, but strongly distorted in others." (iii) Pennington and Worah, *NeurIPS* 2018, "The spectrum of the Fisher information matrix of a single-hidden-layer neural network," derive the exact Fisher spectrum in the infinite-width limit and observe that "linear networks suffer worse conditioning than any nonlinear network, and although nonlinear networks may have many small eigenvalues they are generically non-degenerate." (iv) The PNAS 2024 paper "The training process of many deep networks explores the same low-dimensional manifold," DOI

10.1073/pnas.2310002121, directly connects sloppy-model geometry to the trained-network manifold. (arXiv +2 + 3)

For the manuscript's headline empirical claim about MLIP error covariance specifically, I will cite Subramanyam and Perez's information-entropy-based active learning analysis (referenced in arXiv:2502.21165) and the Bartók-Kermode GPR-UQ work (arXiv:2206.08744) as documenting that MLIP residuals lie in low-dimensional subspaces of the descriptor space. The "Comparing the latent features of universal machine-learning interatomic potentials" preprint (Chorna, Tisi, Malosso, How, Ceriotti, Chong, arXiv:2512.05717, January 2026) provides the most direct comparison of latent feature spaces across uMLIPs to date.

1C. Cross-paradigm cosine-alignment for specific elements/properties

The systematic-softening result (Deng et al., npj CM 2025) is the strongest cosine-alignment finding currently in the literature: three independent uMLIPs share an error direction in property space (underestimated PES curvature → underestimated vibrational frequencies, surface energies, defect energies, migration barriers). On the molecular side, the MACE-OFF23 paper and the AIMNet2/ANI-2x lineage demonstrate that intermolecular force errors decompose into a class-shared dispersion-region direction and model-specific short-range corrections.

For the manuscript I will not over-claim element-resolved cosine alignment because the published evidence is property-resolved rather than element-resolved. Magnetic systems (Fe, Co, Ni) and heavy elements with relativistic / spin-orbit effects are the clearest a priori candidates for shared structural inadequacy because most foundation MLIPs trained on Materials Project PBE data inherit the same missing physics, but the manuscript will treat this as a pre-registered prediction (clause (iv)/(vi) falsifiability) rather than as an established empirical anchor.

PART 2: CROSS-MODEL VANDERMONDE EXTENSION

2A. The single-model anchor and the gap

Waterfall, Casey, Gutenkunst, Brown, Myers, Brouwer, Elser, Sethna, *Phys. Rev. Lett.* 97, 150601 (2006), DOI 10.1103/PhysRevLett.97.150601 (arXiv:cond-mat/0605387), is the single-model anchor. They show that for a wide class of multiparameter nonlinear models the Hessian of the cost function is approximately of the form $V^T A^T A V$, where V is a Vandermonde matrix, and that the m -th largest eigenvalue λ_m is bounded above by an expression of order $\epsilon^{2(m-1)}$: in their words, "the m -th largest eigenvalue is bounded above by the m -th largest row sum." This produces the canonical sloppy spectrum — geometric eigenvalue decay over many orders of magnitude. The bound is *single-model*: it applies to one parametric family at a time. (Cornell)

The clause-(iii) statement of the Universality Theorem requires a *cross-model* extension. Formally, given two foundation MLIPs M and M' both in the architectural class F , we want to bound the Vandermonde decay rates $\rho(M)$ and $\rho(M')$ by a constant $\rho(F)$ depending only on the class. The published literature does not contain this cross-model statement explicitly. This becomes the manuscript's headline mathematical lemma.

2B. Mathematical machinery needed for the cross-model lemma

Four ingredients suffice.

(i) Concentration of empirical Fisher matrices. Tropp, "User-friendly tail bounds for sums of random matrices," *Foundations of Computational Mathematics* 12, 389–434 (2012), DOI 10.1007/s10208-011-9099-z (arXiv:1004.4389), provides matrix Bernstein and matrix Chernoff inequalities — in Tropp's own words, "noncommutative generalizations of the classical bounds associated with the names Azuma, Bennett, Bernstein, Chernoff, Hoeffding, and McDiarmid." For an empirical Fisher matrix $F_n(M) = (1/n) \sum_i \nabla_{\theta} \ell(x_i; \theta) \nabla_{\theta} \ell(x_i; \theta)^{\top}$ evaluated for foundation MLIP M , Tropp's matrix Bernstein bound gives $\|F_n(M) - F(M)\|_{\text{op}} \leq C \sqrt{(\log d / n)}$ with high probability under bounded-variance assumptions. The constant C depends on the loss-gradient sup-norm, which by (F3) (uniform smoothness) is uniformly bounded across $M \in F$. This is the first step of the proof. [Springer](#) [Springer](#)

(ii) Stability of singular value decay under perturbation. Wedin's $\sin-\theta$ theorem (Wedin, *BIT Numerical Mathematics* 12, 99–111, 1972, DOI 10.1007/BF01932678) states that for matrices A and $A + E$ with singular spectra differing by $\leq \varepsilon$ in operator norm, the singular subspaces rotate by at most ε / gap . Combining matrix Bernstein with Wedin gives: the m -th singular value of the empirical Jacobian $J_n(M)$ is within $O(\sqrt{(\log d / n)})$ of the population value $\sigma_m(J(M))$. I will use the Dopico 2000 cleaner $\sin-\theta$ statement (Dopico, *BIT* 40, 395–403, 2000, DOI 10.1023/A:1022303426500) for the Jacobian-perturbation step because it is directly suited to singular subspaces rather than eigenvectors.

(iii) Lipschitz dependence of singular spectrum on model parameters. Standard Weyl/Hoffman–Wielandt inequalities give $|\sigma_m(J(M)) - \sigma_m(J(M'))| \leq \|J(M) - J(M')\|_{\text{op}}$. The hypothesis (F3) — smoothness of $M(\cdot)$ as a function of atomic positions outside a measure-zero set — implies that the Jacobian map $M \mapsto J(M)$ is Lipschitz on the appropriate sublevel set. Combining this with (F1) (overlapping training distribution) and (F2) (above-incompleteness representation) yields that the Vandermonde decay rate $\rho(M)$ is a Lipschitz functional on F .

(iv) Class-uniform smoothness assumptions. The class F is defined precisely so that (F1)+(F2)+(F3) close the loop. The published anchors for these are: (F1) Materials Project trajectory data (MPtrj), Alexandria/sAlex (Schmidt et al.), OMat24 (Barroso-Luque, Wood, Gao, Dzamba, Fu, Zitnick et al., arXiv:2410.12771), and MAD (Mazitov, Chorna, Fraux et al., *Sci. Data* 12, 1857, 2025); (F2) Pozdnyakov–Willatt–Bartók–Ortner–Csányi–Ceriotti, *PRL* 125, 166001 (2020) for the incompleteness lower bound, together with Joshi–Bodnar–Mathis–Cohen–Liò, ICML 2023 (arXiv:2301.09308), and Nigam–Pozdnyakov–Huguenin–Dumittan–Ceriotti, arXiv:2302.14770,

"Completeness of Atomic Structure Representations," for the architectures that escape the incompleteness floor; (F3) standard smoothness as in MACE, Allegro, NequIP analyses — Bartók-Csányi GAP foundations and the more recent Bigi-Huguenin-Dumittan-Cerioti-Manolopoulos, *J. Chem. Phys.* 157, 234101 (2022), "A smooth basis for atomistic machine learning." [\(APS Journals\)](#)

2C. Statement of the cross-model Vandermonde lemma

In display form, the manuscript will state:

Lemma (Cross-Model Vandermonde, manuscript-original). Let F be the class of foundation MLIPs satisfying (F1), (F2), (F3) with class constants κ_1 (training-distribution coverage), κ_2 (above-incompleteness margin), κ_3 (Jacobian Lipschitz constant). Then there exists $\rho(F) \leq C \kappa_1^{-1} \kappa_2 \kappa_3$ such that for every $M \in F$, the empirical Fisher singular spectrum $\{\sigma_m(F_n(M))\}_{m=1}^d$ satisfies

$$\sigma_m(F_n(M)) \leq \sigma_1(F_n(M)) \cdot \exp(-\rho(F) \cdot (m-1)) + O_p(\sqrt{(\log d / n)})$$

uniformly over $M \in F$, with the second term controlled by Tropp's matrix Bernstein bound.

The proof outline: by Waterfall et al. 2006 the population Fisher of any one $M \in F$ is approximately Vandermonde with decay rate $\rho(M)$; by (F3) the Jacobian Lipschitz constant is uniformly bounded; by (F1)+(F2) the conditioning parameter γ in Beckermann's Vandermonde bound (Beckermann, "The condition number of real Vandermonde, Krylov and positive definite Hankel matrices," *Numer. Math.* 85, 553-577, 2000, DOI 10.1007/PL00005408) — "the Euclidean condition number of any positive definite Hankel matrix of order $n \geq 3$ may be bounded from below by $\gamma^{n-1}/(16n)$ with $\gamma = \exp(4 \cdot \text{Catalan}/\pi)$ " — is uniformly lower-bounded; combining gives a class-uniform $\rho(F)$. The Tropp/Wedin sequence converts population to empirical with the stated rate. The pre-registered numerical threshold is $\rho(F) \geq 1.5$ (in natural log units per index) for the dominant 20 singular values, measurable on any two foundation MLIPs by finite-difference Jacobian computation. [\(Semantic Scholar\)](#)

PART 3: ACTIVE LEARNING AND CURRICULUM LITERATURE

3A. Clause (iv): active learning regret applicable to MLIP residual fitting

The clause (iv) statement of the Universality Theorem connects projection norms onto the per-model error manifold $M(M)$ to active learning regret bounds. The cleanest theoretical anchor is Raj and Bach, "Convergence of Uncertainty Sampling for Active Learning," ICML 2022 (PMLR Vol. 162), arXiv:2110.15784, which gives non-asymptotic convergence rates (the manuscript wording should be "non-asymptotic excess-risk bound" not "regret bound" to be technically accurate) under linearly separable / low-noise assumptions and extends them to the noisy case. Their $O(d)$ per-iteration cost makes the bound usable for foundation-MLIP descriptor dimensions.

For Gaussian-process active learning, the canonical anchor is Wang and Jegelka, ICML 2017, "Max-value entropy search for efficient Bayesian optimization," together with Kanagawa, Hennig, Sejdinovic, Sriperumbudur, arXiv:1807.02582, "Gaussian Processes and Kernel Methods: A Review on Connections and Equivalences," which gives the kernel-equivalence machinery needed to translate GP-based regret bounds into MLIP descriptor-space bounds.

The MLIP-specific active learning anchors are: Smith, Nebgen, Lubbers, Isayev, Roitberg, "Less is more: Sampling chemical space with active learning," *J. Chem. Phys.* 148, 241733 (2018), DOI 10.1063/1.5023802, who showed that "the AL-based potentials perform as well as the ANI-1 potential on COMP6 with only 10% of the data, and vastly outperforms ANI-1 with 25% the amount of data" — this is the empirical signature of the projection-norm-active-learning relation; Bartók and Kermode, arXiv:2206.08744, for GPR-based UQ for interatomic potentials; Perez, Subramanyam, Maliyov, Swinburne arXiv:2502.07104 (already cited above) for misspecified-model active learning; Lysogorskiy, Bochkarev, Mrovec, Drautz, *Phys. Rev. Materials* 7, 043801 (2023), for active learning strategies for ACE models; and the most recent anchor Willow, Park, Sim, Moon, Min, Yang, Kim, Lee, Myung, "Bayesian E(3)-Equivariant Interatomic Potential with Iterative Restratification of Many-body Message Passing," arXiv:2510.03046 (2025), which introduces a joint energy-force NLL loss yielding "superior accuracy compared to conventional NLL losses" and benchmark improvements on rMD17 (the per-table improvements should be cited from the paper's tables when the manuscript figure is drafted, rather than from any aggregate summary statistic). (arXiv + 2)

The clause (iv) pre-registration threshold: the manuscript will pre-register that for any $M \in \mathcal{F}$, the active-learning excess-risk after T queries scales as $O(d \log N / T)$ where d is the bounded intrinsic dimension of $M(M)$ and N is the configuration space dimension, with the constant matching the Raj-Bach non-asymptotic rate to within a factor of 4. Failure of this threshold across two independently sampled foundation MLIPs would falsify clause (iv).

3B. Clause (vi): stable evolution under foundation MLIP generational succession

The neural scaling laws literature provides the theoretical anchor. Bahri, Dyer, Kaplan, Lee, Sharma, "Explaining neural scaling laws," *PNAS* 121, e2311878121 (2024), DOI 10.1073/pnas.2311878121 (arXiv:2102.06701), provides the "four scaling regimes" (variance-limited vs. resolution-limited in both data and parameters) and explicitly connects resolution-limited scaling to the intrinsic data-manifold dimension — directly relevant to clause (vi). Bordelon, Atanasov, Pehlevan, "A Dynamical Model of Neural Scaling Laws," ICML 2024 (arXiv:2402.01092), gives the kernel-dynamical-mean-field-theory treatment with compute-optimal exponent predictions. Bordelon, Atanasov, Pehlevan, "How Feature Learning Can Improve Neural Scaling Laws," NeurIPS 2024 (arXiv:2409.17858), extends to feature-learning regimes — directly relevant because foundation MLIPs are demonstrably in the feature-learning rather than lazy regime. (arXiv) (arXiv)

The MLIP-specific generational data: MACE-MP-0 (Batatia et al., arXiv:2401.00096, 2024) → MACE-MPA-0, which the ACESuit/mace-foundations release notes describe as "a model trained on an enlarged dataset containing 3.5M new crystals, obtained by combining the MPtraj and sAlex datasets"; the MACE-OFF23/24 molecular family; GNoME first to second generation; MatterSim v1 (arXiv:2405.04967, 2024); OMat24 / UMA scaling (Barroso-Luque, Fu, Wood et al., arXiv:2410.12771, 2024); and the cross-generation phonon-readiness study Loew, Sun, Wang, Botti, Marques, *npj Computational Materials* 11, 178 (2025). The Hänseroth, Flötotto, Qaisrani, Dreßler paper "Fine-Tuning Unifies Foundational Machine-learned Interatomic Potential Architectures at ab initio Accuracy" (arXiv:2511.05337, 2025) reports that "fine-tuning universally enhances force predictions by factors of 5–15 and improves energy accuracy by 2–4 orders of magnitude" across MACE, GRACE, SevenNet, MatterSim, and Orb — the cleanest current evidence that the error manifold structure is stable under generational succession but with model-specific evolution rates. ([GitHub](#)) ([arxiv](#))

The clause (vi) pre-registration: the Matbench Discovery leaderboard provides the falsifiability instrument. The manuscript pre-registers that for any two consecutive generations of a foundation MLIP family in F , the dominant error directions (top-5 PCA components of cross-configuration residuals) overlap with cosine similarity ≥ 0.7 . Two of the documented generational transitions (MACE-MP-0 → MACE-MPA-0; MatterSim-v0 → MatterSim-v1; M3GNet → CHGNet for the Materials-Project family) can be tested directly with public test sets.

3C. Δ -learning error reduction

The Universality Theorem's three escape classes (paradigm-agnostic, paradigm-specific, structural inadequacy) connect to multi-fidelity / Δ -learning theory. The canonical multi-fidelity anchors are Kennedy and O'Hagan, *Biometrika* 87, 1–13 (2001), and Le Gratiet and Garnier, *Int. J. Uncertainty Quantif.* 4, 365–386 (2014), giving the Gaussian-process-cokriging framework. For chemistry-specific Δ -learning the anchors are Bogojeski, Vogt-Maranto, Tuckerman, Müller, Burke, *Nat. Commun.* 11, 5223 (2020); Patra, Batra, Chandrasekaran, Kim, Huan, Ramprasad multi-fidelity work (*npj Comput Mater* lineage); and for foundation MLIPs specifically Huang, Deng, Zhong, Kaplan, Persson, Ceder, arXiv:2504.05565 (2025), which provides the empirical Δ -learning scaling law across functionals; and Christiansen and Hammer, arXiv:2502.21179 (2025), which provides the GPR-on-MLIP-embedding Δ -learning machinery — the closest published analogue to the corollary structure the Universality Theorem requires. ([arXiv](#))

No formal theorem currently bounds Δ -learning error from above in terms of base-model error manifold structure. This is the second manuscript-original mathematical result (after the cross-model Vandermonde lemma): the manuscript will state a Δ -learning corollary giving an upper bound on residual error after correction by a GPR on the base MLIP's embedding, in terms of (a) the projection of the residual onto the per-model error manifold $M(M)$ and (b) the dimension $d(F)$. The proof uses Niyogi–Smale–Weinberger sample complexity (Part 4A) plus the cross-model Vandermonde lemma (Part 2C).

PART 4: FOUNDATIONAL PRIMITIVES VERIFICATION

4A. Niyogi-Smale-Weinberger 2008

Niyogi, Smale, Weinberger, "Finding the homology of submanifolds with high confidence from random samples," *Discrete & Computational Geometry* 39, 419–441 (2008), DOI 10.1007/s00454-008-9053-2, is the direct primitive for clause (i)–(ii). The 2019–2026 refinements relevant to the manuscript are: Aamari and Levrard, "Non-asymptotic rates for manifold, tangent space, and curvature estimation," *Annals of Statistics* 47, 177–204 (2019), DOI 10.1214/18-AOS1685, which gives the optimal minimax rates for tangent-space and second-fundamental-form estimation; Aamari, Berenfeld, Levrard, "Optimal reach estimation and metric learning," *Annals of Statistics* 51, 1086–1108 (2023), DOI 10.1214/23-AOS2281, which closes the gap on reach estimation; Aamari, Kim, Chazal, Michel, Rinaldo, Wasserman, *Electron. J. Statist.* 13, 1359–1399 (2019), on tangent-space estimation; Berenfeld, Harvey, Hoffmann, Shankar, "Estimating the reach via convexity defect function," arXiv:2001.08006, published *Discrete Comput. Geom.* 67, 403–438 (2022), DOI 10.1007/s00454-021-00290-8; Attali, Kouřimská, Fillmore, Ghosh, Lieutier, Stephenson, Wintraecken, arXiv:2206.10485 (2022), "Tight bounds for the learning of homotopy à la Niyogi, Smale, and Weinberger," which gives explicit tight bounds for one-sided Hausdorff sampling — directly relevant because the MLIP residual sample is a one-sided Hausdorff sample of the underlying error manifold. (Springer + 3)

For the manuscript: the sample complexity $O(d \log N)$ in clause (ii) is the Niyogi-Smale-Weinberger bound under the reach condition. The 2019/2022/2023 refinements collectively justify using this bound for the noisy / one-sided-Hausdorff regime appropriate to MLIP residuals.

4B. Federer 1959

Federer, "Curvature measures," *Trans. Amer. Math. Soc.* 93, 418–491 (1959), DOI 10.1090/S0002-9947-1959-0110078-1, is the canonical reach reference. The Lipschitz statement needed by the manuscript — that the nearest-point projection ξ_A onto a set A of reach r , restricted to the q -tubular neighborhood with $q < r$, is Lipschitz with constant $r/(r - q)$ — is Federer's Theorem 4.8(8). The companion paper Foote, "Regularity of the distance function," *Proc. Amer. Math. Soc.* 92, 153–155 (1984), DOI 10.1090/S0002-9939-1984-0749908-9, gives the $C^{\{1,1\}}$ structure of distance level sets. For noisy / 2020–2026 settings, Berenfeld and Hoffmann ("Estimating the manifold's reach"), arXiv:2001.08006, and the Aamari-Berenfeld-Levrard 2023 *Annals of Statistics* paper extend Federer's machinery to the statistical setting needed for MLIP residual error manifolds.

4C. Machta-Chachra-Transtrum-Sethna 2013

Machta, Chachra, Transtrum, Sethna, "Parameter Space Compression Underlies Emergent Theories and Predictive Models," *Science* 342, 604–607 (1 November 2013), DOI

10.1126/science.1238723 (arXiv:1303.6738), is the canonical "parameter space compression" reference. In their words: "We trace the emergence of an effective theory for long-scale observables to a compression of the parameter space quantified by the eigenvalues of the Fisher Information Matrix. A similar compression appears ubiquitously in models taken from diverse areas of science, suggesting that the parameter space structure underlying effective continuum and universal theories in physics also permits predictive modeling [more broadly]." The natural 2020–2026 extension is Quinn, Abbott, Transtrum, Machta, Sethna, "Information geometry for multiparameter models: New perspectives on the origin of simplicity," *Rep. Prog. Phys.* 86, 035901 (2023), DOI 10.1088/1361-6633/aca6f8 (arXiv:2111.07176), which gives the full information-geometric treatment including the manifold-boundary approximation. For neural networks specifically, the extensions are Karakida–Akaho–Amari AISTATS 2019, Pennington–Worah NeurIPS 2018, and the 2024 PNAS paper "The training process of many deep networks explores the same low-dimensional manifold" (DOI 10.1073/pnas.2310002121) — collectively documenting that the Machta et al. 2013 compression phenomenon survives the transition from physical / biological multiparameter models to over-parametrized neural networks.

(BYU Physics + 3)

4D. Pozdnyakov–Willatt–Bartók–Ortner–Csányi–Ceriotti 2020

Pozdnyakov, Willatt, Bartók, Ortner, Csányi, Ceriotti, *Phys. Rev. Lett.* 125, 166001 (2020), DOI 10.1103/PhysRevLett.125.166001 (arXiv:2001.11696), is the canonical incompleteness floor: three- and four-body descriptors fail to uniquely specify atomic environments. The 2022–2026 clarifications: (APS Journals) (APS Journals)

- Pozdnyakov and Ceriotti, *Mach. Learn. Sci. Technol.* 3, 045020 (2022), "Incompleteness of graph neural networks for atomic structures," which extends the incompleteness result to GNN architectures and shows which message-passing depths suffice.
- Nigam, Pozdnyakov, Huguenin-Dumittan, Ceriotti, "Completeness of Atomic Structure Representations," arXiv:2302.14770 (2023), proves which architectures escape the incompleteness floor.
- Joshi, Bodnar, Mathis, Cohen, Liò, "On the Expressive Power of Geometric Graph Neural Networks," ICML 2023 (PMLR 202:15330–15355), arXiv:2301.09308, develops geometric Weisfeiler–Leman (GWL) and proves the expressiveness hierarchy: equivariant > invariant, higher tensor order > scalar, higher body order > pairwise. (Towards Data Science)
- Bigi, Huguenin-Dumittan, Ceriotti, Manolopoulos, "A smooth basis for atomistic machine learning," *J. Chem. Phys.* 157, 234101 (2022), DOI 10.1063/5.0124363, provides the smoothness anchor for (F3).

For the manuscript, condition (F2) is operationalized as: M lies strictly above the Pozdnyakov–Ceriotti incompleteness floor in the GWL hierarchy of Joshi et al. — i.e., is at least an equivariant message-passing network of depth ≥ 2 with body-order ≥ 4 features (MACE, NequIP, Allegro, eqV2, GRACE, eSEN, PET, SevenNet, Orb-v3, MatterSim, DPA3 all qualify).

4E. Waterfall et al. 2006 and the Vandermonde primitives

Waterfall et al. 2006 (already cited) is the single-model Vandermonde anchor. Beckermann, "The condition number of real Vandermonde, Krylov and positive definite Hankel matrices," *Numerische Mathematik* 85, 553–577 (2000), DOI 10.1007/PL00005408, provides the rigorous lower bound $\gamma^{n-1}/(16n)$ on the Euclidean condition number with $\gamma = \exp(4 \cdot \text{Catalan}/\pi)$. This is the rigorous Vandermonde primitive that the cross-model lemma will lean on. No stronger published Vandermonde result has appeared that supersedes Beckermann 2000 for the bounds the manuscript needs.

PART 5: MANUSCRIPT STRUCTURE PROPOSAL

5A. Target journals and positioning

Primary target. *npj Computational Materials* (Springer Nature). Recent precedent for theoretical-cum-empirical foundation-MLIP papers in this venue: Perez-Subramanyam-Maliyov-Swinburne 2025 (UQ for misspecified MLIPs); Deng et al. 2025 (systematic softening); Loew et al. 2025 (phonons readiness); Bilbrey-Firoz-Lee-Choudhury 2025 (UQ for neural-network potential foundation models). Scope match is excellent. Audience overlap with the MLIP community is maximal. Word budget: standard 5000–7000 words for the main text plus supplementary material for the proofs.

Stretch target. *Nature Computational Science*. The theorem framing and the meta-theorem positioning over CMET justifies the broader-audience venue, but the format constraints (3000–4500 words main text) require moving most proofs to supplementary information.

5B. Section structure with approximate word counts

Title (working). A Universality Theorem for Foundation Machine Learning Interatomic Potentials: Per-Model Error Manifolds with Class-Uniform Geometry.

Abstract (200 words). State the six clauses crisply. Lead with the cross-model Vandermonde lemma as the headline mathematical result. Position the manuscript as a meta-theorem over CMET. Close with the two pre-registered falsifiable predictions.

§1 Introduction (1000 words). Motivate the question by reviewing the foundation MLIP landscape — MACE-MP-0/MPA-0, CHGNet, Orb, SevenNet, eqV2, GNoME, MatterSim, PET-MAD, eSEN, GRACE, DPA3, OMat24/UMA — and the documented heterogeneity of their error structures (Matbench Discovery, MLIP Arena). State the central question: is the heterogeneous error structure model-specific noise, or does it have shared class-level geometry? Preview the six clauses and the falsifiability section. Position the manuscript as the second of a planned trilogy, with the first paper (CMET) providing the manifold-extension primitive and the third paper

(forthcoming) treating refusal-mode behavior. Key citations: Batatia 2024 (MACE-MP-0); Riebesell 2025 (Matbench Discovery); Chiang 2025 (MLIP Arena); Mazitov 2025 (PET-MAD); Deng 2025 (systematic softening).

§2 The Architectural Class F (700 words). Define (F1), (F2), (F3) operationally. (F1) anchored to Materials Project / MPtrj / Alexandria / OMat24 / MAD; (F2) anchored to the Pozdnyakov-Cerioti incompleteness floor and the Joshi et al. GWL hierarchy; (F3) anchored to the Bigi-Huguenin-Dumittan-Cerioti-Manolopoulos smooth basis and the standard equivariant-message-passing smoothness arguments. Worked examples: MACE-MP-0 $\in$ F, MACE-MPA-0 $\in$ F, CHGNet $\in$ F (boundary case, marginal F2 satisfaction), Orb-v3 $\in$ F, MatterSim $\in$ F, eqV2/UMA $\in$ F, PET-MAD $\in$ F, SevenNet $\in$ F, GRACE $\in$ F, eSEN $\in$ F. Counterexamples: classical potentials (fail F2); GAP-original (fails F1 at scale); incomplete-descriptor models (fail F2).

§3 Statement of the Universality Theorem (500 words). Six clauses in display form. Clauses (i)–(iii), (v) are rigorous; clauses (iv) and (vi) are pre-registered predictions with explicit numerical thresholds.

§4 Empirical Foundations (1200 words). Three subsections: (4.1) cross-model heterogeneity from Matbench Discovery, MLIP Arena, MOFSimBench (Nature 2025), MDR phonons (Loew 2025), and surface energetics (Focassio 2025); (4.2) sloppy-spectrum structure from Perez-Subramanyam-Maliyov-Swinburne 2025, Karakida-Akaho-Amari 2019, Sagun et al. 2017, Pennington-Worah 2018, Quinn-Abbott-Transtrum-Machta-Sethna 2023, and the PNAS 2024 manifold paper; (4.3) cross-paradigm cosine alignment from Deng et al. 2025 (systematic softening) and Christiansen-Hammer 2025 (Δ -correction).

§5 Rigorous Proofs of Clauses (i)–(iii), (v) (1500 words). §5.1 clause (i) intrinsic dimension via Niyogi-Smale-Weinberger 2008 plus Aamari-Levrard 2019 refinements; §5.2 clause (ii) sample complexity $O(d \log N)$ directly from Niyogi-Smale-Weinberger plus the Attali et al. 2022 one-sided Hausdorff tightening; §5.3 clause (iii) cross-model Vandermonde lemma — proof sketch in the main text, full proof in Appendix A; §5.4 clause (v) two-mode inference / refusal-region characterization via Federer 1959 reach machinery plus the Berenfeld-Hoffmann-Shankar 2020/2022 noisy-reach extension. Worked example: surface-energy refusal mode (cf. Focassio et al. 2025).

§6 Pre-Registered Predictions for Clauses (iv) and (vi) (700 words). §6.1 active learning excess risk (clause iv): pre-register $O(d \log N / T)$ scaling with constant matching Raj-Bach 2022 to within factor 4, tested on PET-MAD and MACE-MPA-0 against a held-out chemically-diverse test set. §6.2 generational stability (clause vi): pre-register top-5 PCA error-direction cosine similarity ≥ 0.7 between consecutive generations, testable on MACE-MP-0 $\rightarrow$ MACE-MPA-0, MatterSim-v0 $\rightarrow$ MatterSim-v1, M3GNet $\rightarrow$ CHGNet. Each prediction explicitly stated with falsification threshold.

§7 Corollary: Three Escape Classes (500 words). Paradigm-agnostic (errors in span of the class-uniform component, addressable by Δ -learning à la Christiansen–Hammer); paradigm-specific (errors in the model-specific component, addressable by fine-tuning à la Hänseroth et al. 2025); structural inadequacy (errors in the refusal region, requires architectural change). Each escape class cited to a published Δ -correction or fine-tuning result. Connection to multi-fidelity literature (Huang et al. 2025).

§8 Connection to CMET (400 words). State the meta-theorem framing: CMET is the manifold-extension primitive for a single MLIP; the Universality Theorem promotes it to a class-level statement. Defer detailed proof of the CMET reduction to Appendix D.

§9 Discussion and Falsifiability Summary (500 words). Restate both pre-registered predictions. Itemize three open mathematical problems: (1) the class-uniform smoothness constant κ_3 in (F3), currently bounded by GAP-style smoothness analyses but not by a published constant for equivariant-message-passing architectures; (2) the sharpness of the cross-model Vandermonde lemma — is the constant $\rho(F)$ tight?; (3) the connection between refusal region and the Pozdnyakov–Ceriotti incompleteness manifold.

Appendices. (A) Cross-Model Vandermonde Lemma proof — full version, ~1500 words plus equations; (B) Architectural class F definition with full worked examples including the boundary cases CHGNet, M3GNet, and the non-F counterexample of classical EAM potentials; (C) Refusal-region characterization with the surface-energy worked example; (D) Connection to CMET.

Total main-text target ~7000 words, fitting the npj Computational Materials format with extensive appendices in supplementary information.

5C. Key citations per section

- §1 Introduction: Batatia 2024, Riebesell 2025, Chiang 2025, Mazitov 2025, Deng 2025, Huang 2025, Loew 2025, Pozdnyakov 2020.
- §2 Class F: Batatia 2024, Bochkarev 2024, Rhodes 2025, Yang 2024, Mazitov 2025, Pozdnyakov–Ceriotti 2020/2022, Nigam 2023, Joshi 2023, Bigi 2022.
- §4 Empirical: Riebesell 2025, Chiang 2025, Mazitov 2025, Loew 2025, Focassio 2025, Deng 2025, Christiansen–Hammer 2025, Perez 2025, Karakida 2019, Sagun 2017, Pennington–Worah 2018, Quinn 2023.
- §5 Proofs: Niyogi–Smale–Weinberger 2008, Federer 1959, Aamari–Levrard 2019, Aamari–Berenfeld–Levrard 2023, Attali 2022, Waterfall 2006, Beckermann 2000, Tropp 2012, Wedin 1972, Dopico 2000, Foote 1984.
- §6 Predictions: Raj–Bach 2022, Smith 2018, Willow 2025, Bahri 2024, Bordelon 2024a/b, Hänseroth 2025.
- §7 Escape classes: Christiansen–Hammer 2025, Hänseroth 2025, Huang 2025, Bogojeski 2020, Kennedy–O’Hagan 2001, Le Gratiet–Garnier 2014.

5D. Manuscript-original mathematical content and honesty about literature gaps

Three results are manuscript-original: (i) the cross-model Vandermonde lemma in Part 2; (ii) the Δ -learning corollary in Part 3C; (iii) the formal definition and structure of the class F as a closure of $(F1)+(F2)+(F3)$. The pre-registered predictions in §6 are testable predictions, not theorems.

Areas of the public literature that are thin:

- **MLIP-specific Fisher spectrum.** No published paper directly computes the empirical Fisher of a MACE-MP-0-scale foundation MLIP and shows the sloppy spectrum. Perez-Subramanyam-Maliyov-Swinburne 2025 is the closest, but uses parameter covariance rather than Fisher spectrum. The manuscript should commission this computation as a §4 figure.
- **Cross-model cosine alignment for elements.** Deng et al. 2025 provides cross-paradigm property-direction alignment; nothing comparable exists element-by-element. The manuscript will treat element-resolved cosine alignment as a pre-registered (clause iv/vi) prediction rather than as established empirics.
- **Cross-model Vandermonde.** As noted, this is manuscript-original and constitutes the headline mathematical result.

Areas where the literature is dense and direct citations suffice:

- Niyogi-Smale-Weinberger sample complexity and its 2019–2023 refinements.
- Federer reach machinery and its noisy-statistical refinements.
- Pozdnyakov-Cerioti incompleteness and the geometric-GNN expressiveness hierarchy (Joshi et al. 2023, Nigam et al. 2023).
- Sloppy-model universality (Waterfall 2006, Quinn 2023).
- Neural scaling laws (Bahri 2024, Bordelon 2024a/b).

5E. Pre-registration anchors and falsifiability

The manuscript closes with an explicit falsifiability section. Two pre-registered predictions:

P1 (clause iv). For any two foundation MLIPs $M, M' \in F$ evaluated on a chemically diverse held-out test set, the active-learning excess risk after T queries decays as $\epsilon(T) = C \cdot d \log N / T$ with d the empirical intrinsic dimension of $M(M)$ (measured by local PCA on residuals) and C within a factor of 4 of the Raj-Bach 2022 constant. Falsification: if either C or the scaling exponent in T differs by more than the stated factor, the Universality Theorem clause (iv) is falsified for that pair (M, M') .

P2 (clause vi). For any foundation MLIP family with two consecutive generations M_g and M_{g+1} , the top-5 principal directions of the cross-configuration residual matrix have pairwise

cosine similarity ≥ 0.7 with the top-5 principal directions of the next generation, evaluated on the Matbench Discovery WBM test set or equivalent. Falsification: cosine similarity below 0.5 for any consecutive generational transition within F falsifies clause (vi).

The falsifiability section ties these predictions to publicly accessible test sets and to the Matbench Discovery / MLIP Arena infrastructure so that any reader can run the test independently.

SUMMARY AND HANDOFF NOTES FOR MANUSCRIPT WRITING

The public-literature replacement is complete for the empirical anchors (§4), the foundational primitives (§5 proofs of clauses i, ii, v), and the active-learning / scaling-law backdrop (§6 pre-registration). The single piece of manuscript-original mathematical content — the cross-model Vandermonde lemma — assembles from four published primitives (Waterfall 2006 single-model Vandermonde; Tropp 2012 matrix Bernstein; Wedin 1972 / Dopico 2000 $\sin-\theta$; Beckermann 2000 Vandermonde conditioning) plus the class-F definition. The Δ -learning corollary in §7 assembles from Niyogi-Smale-Weinberger sample complexity plus the cross-model Vandermonde lemma. No internal-corpus citations remain in this preparation document; every cited fact is anchored to an arXiv ID, DOI, or benchmark URL accessible to the reader.

The next steps for manuscript writing are: (1) commission the empirical MLIP-Fisher computation for §4 (one figure showing the sloppy spectrum for MACE-MP-0 and PET-MAD); (2) write out the full cross-model Vandermonde proof in Appendix A using the Tropp/Wedin/Beckermann sequence; (3) write out the Δ -learning corollary in §7 using Niyogi-Smale-Weinberger plus the lemma from Appendix A; (4) pre-register predictions P1 and P2 with explicit test-set protocols on Matbench Discovery and MLIP Arena. Recommended initial submission target: npj Computational Materials, with the title and abstract leading on the cross-model Vandermonde lemma and the two pre-registered predictions.